

# GADIPALLY VINHU PPRAWIN

Mechanical Design Engineer | CAD & Structures | Robotics & Automation | Aerospace Manufacturing

Hyderabad, Telangana / Bengaluru, Karnataka | +91 6300142746 | [vinhupraveen@gmail.com](mailto:vinhupraveen@gmail.com) | [linkedin.com/in/contactvinhu](https://www.linkedin.com/in/contactvinhu) | [gvinhupraveen.github.io](https://github.com/gvinhupraveen)

## PROFESSIONAL SUMMARY

Mechanical Engineer with hands-on experience across CAD/CAE (CATIA, Siemens NX, SolidWorks, ANSYS), composite structures, and test instrumentation (NI-DAQ, LabVIEW), worked at work at Hindustan Aeronautics Limited aerospace MRO, precision CNC manufacturing, and current work supporting industrial 3D-printing systems at Fracktal Works. Independently designed, fabricated, and validated an ASTM D7136-compliant low-velocity impact test rig with a custom NI-DAQ/LabVIEW data acquisition system for composite specimen testing. Led structural design and manufacturing for MIT Manipal's AeroMIT team to a 4th-place world ranking at the SAE Aero Design Competition.

## TECHNICAL SKILLS

- **CAD/CAE & Mechanical Design:** CATIA • Siemens NX • SolidWorks • Fusion 360 • AutoCAD • ANSYS Static Structural • Assembly Modeling • GD&T • Engineering Drawings
- **Instrumentation, DAQ & Test:** LabVIEW • NI-DAQ • Signal Conditioning • Sensor Integration (Load Cell) • High Sampling-Rate Data Acquisition • Trigger Logic • ASTM D7136 Impact Testing
- **Structures & Materials:** Structural Design • Composite Materials (Carbon Fibre/GFRP Layup) • Material Characterisation • Load Analysis • Weight Optimisation
- **Manufacturing & Production:** VMC/CNC Operation • G-Code Programming • Manual Lathe & Milling • Turning, Threading, Fitting & Bench Work • Servo & Pneumatic System Integration
- **Quality & Aerospace Standards:** AS9100 Awareness • DGCA-Aligned MRO Practice • FOD Prevention • Dimensional Metrology • SPC • IFS Cloud ERP
- **Hardware, Firmware & 3D Printing:** FDM & Resin (SLA/MSLA) 3D Printing • Machine Calibration • Klipper Firmware Configuration • BIGTREE TECH Controller Flashing • Multi-MCU/CAN-Bus Setup
- **Documentation & Process:** Technical Documentation • Root Cause Analysis • CAPA • Cross-Functional Collaboration • Team Leadership & Mentoring

## PROFESSIONAL EXPERIENCE

### Technical Support Engineer

Fracktal Works Pvt Ltd | Bengaluru, Karnataka | May 2026 – Present

- Provide technical support for Fracktal's professional FDM and resin 3D printing systems, diagnosing hardware, firmware, and software issues to minimise downtime for aerospace, automotive, and product-design clients.
- Diagnose and resolve mainboard, firmware, and calibration issues across the printer fleet, including Klipper-based multi-MCU/CAN-bus configurations and BIGTREE TECH controller firmware flashing.
- Collaborate with engineering and product teams to troubleshoot print failures, analyse root causes, and implement corrective actions on print parameters, slicing settings, and mechanical components.
- Support installation, commissioning, and maintenance of 3D printing equipment at client sites; document service reports and contribute to the technical knowledge base.

### Graduate Apprentice Trainee – Quality Assurance

Hindustan Aeronautics Limited (HAL), MRO Division | Bengaluru, Karnataka | Jan 2026 – Apr 2026

- Trained within HAL's Maintenance, Repair & Overhaul (MRO) division for helicopter transmission systems, applying DGCA and AS9100 aerospace quality frameworks to assembly of the Main Gear Box (MGB) of the ALH Dhruv helicopter.
- Reviewed technical documentation — engineering drawings, inspection records, non-conformance reports — using IFS Cloud ERP.
- Performed precision measurement, tolerance verification, and dimensional inspection of critical rotating components in a live aerospace manufacturing environment; built strong shop-floor discipline and FOD-prevention practice.

### Quality Trainee – Plastic Body Division

Tata Motors Autocomp Plastic Division | Sanand, Gujarat | Nov 2025 – Jan 2026

- Monitored shop-floor production quality for Tata Sierra and Nexon door-body components, applying visual inspection, gauge-based measurement, and SPC to identify and escalate non-conformances.
- Coordinated with production supervisors on corrective and preventive actions (CAPA) for recurring quality issues in a high-volume automotive manufacturing environment.

### VMC Operator & CAD Designer

Avinya Engineering Solutions | Pune, Maharashtra | Jul 2025 – Nov 2025

- Operated a Vertical Machining Centre (VMC) to machine aerospace and automotive components to tight tolerances per engineering drawings.
- Designed and built a custom automatic tapping machine: integrated servo-controlled linear motion of the tapping head with a pneumatically actuated ejection cylinder, reducing manual intervention and cycle time for a repetitive tapping operation.
- Created 2D drawings and 3D models in SolidWorks and Fusion 360 for machined components and custom automation fixtures.

## KEY PROJECTS

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### Low-Velocity Drop-Weight Impact Test Setup — Final-Year B.E. Project

MIT Manipal | Departmentally funded, 6-month project | Sole designer & builder | 2025

- Owned the full mechanical design — frame layout, guided drop mechanism, specimen fixture — meeting ASTM D7136 requirements, with controlled impact energy via adjustable mass and drop height.
- Designed and optimised a NI-DAQ/LabVIEW data acquisition system: captured impact force–time histories from a load cell at high sampling rates, tuning signal filtering and trigger logic for reliable peak-force detection on short-duration events.
- Iterated through CAD/FEA design cycles and hands-on assembly trials to resolve alignment, stability, and repeatability issues; validated the rig across multiple runs on composite laminates.
- The rig is now an active departmental facility supporting ongoing composite material characterisation and research.

### Structures Head — AeroMIT, MIT Manipal

SAE International Aero Design Competition, Florida, USA | 2023–2024

- Led the structures sub-team of AeroMIT, MIT Manipal's aeromodelling and aerial robotics team, owning structural design, material selection, and manufacturing for competition UAVs.
- Achieved Overall World Rank 4 and World Rank 6 (2023–24) against international university teams, and World Rank 2 in the Design Report category (2022–23).
- Trained junior members in composite layup, weight optimisation, and load analysis for UAV airframes.

### Custom Cartesian 3D Printer — Independent Project

- Designed, sourced, and assembled a fully functional Bowden-drive Cartesian 3D printer from scratch — aluminium-extrusion frame, linear motion system (rods, lead screws, lead nuts), and extruder assembly.
- Performed electronics integration, firmware configuration, and iterative calibration to reach print-ready, repeatable performance.

## EDUCATION

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- **M.Tech – Robotics and Artificial Intelligence (Hybrid)** | IIT Guwahati | Commencing Aug 2026
- **B.Tech – Mechanical Engineering (Manufacturing)** | Manipal Institute of Technology, Udupi, Karnataka | 2021–2025
- **Diploma – Mechanical Engineering** | Pallavi Engineering College, Telangana | 2019–2022 | 83.7%
- Industrial Training — Central Institute of Tool Design (CITD), MSME Tool Room, Hyderabad (6 months): manual lathe/milling, CNC/VMC G-code programming, GD&T, precision measurement.

## ACHIEVEMENTS & CERTIFICATIONS

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- SAE International Aero Design Competition: Overall World Rank 4 (2023–24), World Rank 6 (2023–24), Design Report Rank 2 (2022–23), Florida, USA.
- Departmentally funded final-year project: designed and delivered an ASTM D7136 low-velocity impact test facility now in active departmental use.
- Independently designed, built, and calibrated a fully functional Cartesian 3D printer end-to-end, including firmware and electronics integration.

## ADDITIONAL INFORMATION

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Languages: English (Professional) • Telugu (Native) • Hindi (Conversational) • Kannada (Basic)

Interests: UAV/RC Aircraft Design • 3D Printing & Rapid Prototyping • Topology Optimisation • Robotics & Automation

Location: Currently based in Bengaluru; home base Hyderabad. Open to relocation.